

Inputs: External dims.
 $(\Delta_{ij}, \Delta_{kj})$; evaluation points
 $\{z_1, \dots, z_N\}$; spacetime dim. D ;
operator data (Δ_a, ℓ_a) ;
block types (F_+, F_-)

Phase 1 — Recursion

Per z_i (parallel workers)

Setup: compute parameters,
convert $z_i \rightarrow (r, \eta)$ and $1-z_i \rightarrow (r_1, \eta_1)$

Determine recursion poles:
compute pole positions, coefficients, and residues for each spin

Seed values:
evaluate $\tilde{h}_\ell^{\Delta_{ij}, \Delta_{kl}}(r, \eta)$ at each (r, η)

Block iteration:
Calculate successive h_{new} for previous h (dependence on operators suppressed for simplicity).

Converged?

No

No

max_iter
hit?

Yes

Yes

Phase 2 — Block evaluation

Evaluate g blocks:
for each (Δ_a, ℓ_a) , compute g
at both (r, η) and (r_1, η_1)

Phase 3 — $F_\pm$ construction

Construct $F_\pm$ blocks:
 $F_\pm = v^{\bar{\Delta}_{jk}} g(u, v) \pm u^{\bar{\Delta}_{jk}} g(v, u)$

Synchronisation barrier:
collect results from all workers

Output: $F_{\pm, \Delta_a, \ell_a}(z_i, \bar{z}_i)$ for all z_i